RW–10638

Sub. Code
541561/ 545561

M.C.A.(R)/WE DEGREE EXAMINATION, NOVEMBER 2015

SOFT COMPUTING

(Common for M.C.A.(R)/WE)

(2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. What are the different learning paradigms?

2. What is the main difference between single layer and multilayer Artificial Neural network?

3. List out the various activation functions used in ANN.

4. What is the objective of training the network?

5. Define Fuzzy set.

6. State defuzzification techniques.

7. What is meant by fuzzy measure?

8. What is the main difference between probability and fuzzy logic?

9. What is mutation?

10. What are the elements of Genetic algorithm?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

All questions carry equal marks.

11. (a) Describe the application and scope of neural networks.

Or

(b) Explain the training of Artificial Neural Network.

12. (a) Explain about the Adaline network.

Or

(b) Describe the Kohonen Self organizing Network.

13. (a) Write the differences between Crisp set and Fuzzy set.

Or

(b) Write about Fuzzy relation.

14. (a) Describe the fuzzy measure.

Or

(b) Discuss about Fuzzy Decision making.

15. (a) Describe the Encoding in genetic algorithm.

Or

(b) Briefly explain fitness functions.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

All questions carry equal marks.

16. Explain the architecture of Neural network.

17. Explain Back propagation network in detail.

18. Describe the Defuzzification method.

19. Discuss about Sugeno -style fuzzy inference process.

20. List out the Genetic operators and explain them.

———————

RW–10484

Sub. Code
541102/ 545102

M.C.A. DEGREE EXAMINATION, NOVEMBER 2015

C AND DATA STRUCTURES

(Common for M.C.A. (R)/M.C.A. (WEP))

(2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** the questions.

1. List the rules for defining variables.

2. List out any two differences between do-while and while statements.

3. What is the use of recursion?

4. Define any two string functions.

5. Why do we need data structures?

6. List any four Common data structures.

7. What is the degree of node?

8. Define : Binary tree.

9. List the difficulties of Linear Search.

10. What is the time for Quick Sort?

Section B (5 × 5 = 25)

Answer **all** the questions.

11. (a) Write a short note on data types.

Or

(b) Explain the I/O functions in detail.

12. (a) Explain the arrays in detail.

Or

(b) With suitable example explain the arithmetic operations with pointers.

13. (a) Write a short note on Operations performed on any linear structure.

Or

(b) What is QUEUE data structure? Explain the Operations of Queue.

14. (a) With the help of tree diagram explain the following terms :

(i) Parents

(ii) Children

(iii) Siblings

(iv) Leaves

(v) Length.

Or

(b) Explain the insertion and deletion operations on binary tree.

15. (a) With the suitable example explain the linear search.

Or

(b) Explain the binary search with an example.

Section C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Describe the Control structures in detail.

17. Write a C program to arrange the given numbers in

(a) Ascending Order

(b) Descending Order.

18. Write a C program to implement the following operations on linked list :

(a) CREATION

(b) INSERT AT STARTING

(c) INSERT AT MIDDLE (USER'S CHOICE)

(d) INSERT AT END

(e) DELETE 1ST NODE

(f) DELETE LAST NODE

(g) DELETE MIDDLE NODE (USER'S CHOICE)

(h) DISPLAY.

19. Describe the binary tree traversals in detail.

20. Explain any two sorting techniques with example.

———————

RW–10485

Sub. Code
541103/ 545103

M.C.A. DEGREE EXAMINATION, NOVEMBER 2015

RELATIONAL DATABASE MANAGEMENT SYSTEMS

(Common for M.C.A. (R) / M.C.A. (WEP))

(2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. What are the types of Data Language?
2. What is the use of ER Diagram?
3. What are key constrains?
4. What is tuple relational calculus?
5. List out the Aggregate functions in SQL.
6. What is need of Normalization?
7. What is Atomicity?
8. What is the mechanism of Time Stamp based protocol?
9. How to use indexing in queries?
10. What is cluster Indexes?

Part B (5 × 5 = 25)

Answer **all** the questions.

11. (a) Describe any two data models with diagram.

Or

(b) Explain Storage Manager and its types.

12. (a) Illustrate any two Relational Calculus Operations with example.

Or

(b) Explain Cartesian product with Example.

13. (a) Explain Nested Queries and Aggregate operators with Example.

Or

(b) What is Functional Dependency? Describe with Example.

14. (a) Explain different types of Transaction States.

Or

(b) Describe Buffer Management with Block Diagram.

15. (a) Explain about Indexing and its types.

Or

(b) Describe about B+ trees.

Part C $(3 \times 10 = 30)$

Answer any **three** the questions.

16. Explain the DBMS Architecture with Block Diagram.

17. Describe the Relational Algebra operations in detail.

18. Discuss the normalization in detail.

19. Describe the different types of transaction protocols in detail.

20. Explain different types of DBMS Data Storage with block diagram.

RW–10486

Sub. Code
541104/ 545104

M.C.A. DEGREE EXAMINATION, NOVEMBER 2015

Computer Applications

DISCRETE MATHEMATICS

(Common for M.C.A. (R)/M.C.A. (WEP))

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. What is Tautology? Explain the term contradiction.

2. Construct the Truth table for $(P \vee Q) \vee \rceil P$.

3. Given $A = \{2,3,4\}, B = \{1,2\}, C = \{4,5,6\}$ find (a) $A + B$ (b) $B + C$ (c) $A + B + C$ (d) $(A + B) + (B + C)$.

4. Prove $A \cup A = A$ and $A \cap A = A$.

5. Find all the Subgroups of S4 generated by the permutation $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 3 & 2 & 4 \end{pmatrix}$ and $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix}$.

6. What is called an Algebraic system?

7. What is Graph? Explain weighted graph with a diagram.

8. How many different directed trees are there with three nodes?

9. What is Sample space? Explain with an example.

10. What is the Probability?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Show that $(\rceil P \wedge (\rceil Q \wedge R)) \vee (Q \wedge R) \vee (P \wedge R) \Leftrightarrow R$.

Or

(b) Show that $P \wedge \rceil P \wedge Q \Rightarrow R$ and $R \Rightarrow P \vee \rceil P \vee Q$.

12. (a) Show that for any two sets A and B $A - (A \cap B) = A - B$.

Or

(b) Let E be the identity relation on a set X and R be any relation in X; Show that $E \cup R \cup \widetilde{R}$ is a compatibility relation.

13. (a) Show that among the cosets determined by subgroup S in a group $< G, * >$, only one of the cosets is a subgroup.

Or

(b) Show that every subgroup of a cyclic group is normal.

14. (a) Prove that in an acyclic simple diagraph a node base consist of only those nodes whose in degree is zero.

Or

(b) Show that the sum of in degrees of all the nodes of a simple diagraph is equal to the sum of out degree of all its nodes and that this sum is equal to the number of edges of the graph.

15. (a) Explain Baye's Theorem.

Or

(b) If A and B are independent events then $\overline{A}$ and $\overline{B}$ are also independent events. Prove

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Obtain the Principle conjunctive Normal Form of the formula S given by $(\neg P \rightarrow R) \wedge (Q \rightleftarrows P)$.

17. If $A1 = \{\{1, 2\}, \{3\}\}$, $A2 = \{\{1\}, \{2, 3\}\}$, and $A3 = \{\{1, 2, 3\}\}$, then show that $A1$, $A2$ and $A3$ are mutually disjoint.

18. Show that if $(G, *)$ is a cyclic group then every subgroup of $(G, *)$ must be a cyclic.

19. Show that in a complete binary tree the total number of edges is given by $2(n_t - 1)$, where n_t is the number of terminal nodes.

20. Give a random variable X,

$X = x$	-3	-2	-1	0	1	2	3
$F(x)$	0.1	0.15	0.2	0.05	0.25	0.2	0.05

Find $E(X)$ and standard deviation $\sigma(X)$.

RW–10487

Sub. Code
541105/ 545105/ 551105

M.C.A. DEGREE EXAMINATION, NOVEMBER 2015

Computer Science

SOFTWARE ENGINEERING

(Common for M.C.A. (R)/M.C.A. (WEP)/M.Sc. (CS))

(2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Mention the advantage of layered technology.

2. What is CMMI?

3. Specify the tools used for requirement analysis.

4. Mention the need for data modeling.

5. Mention any four quality attributes.

6. Name any four architectural styles.

7. What is meant by alpha and beta testing?

8. Name any four function-based metrics.

9. Mention the steps involved in risk management.

10. What are risk components?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Explain different categories of software.

Or

(b) Mention the software engineering layers.

12. (a) Write short notes on collaborative requirements gathering.

Or

(b) Mention the advantages of class-based modeling.

13. (a) Explain about architectural patterns.

Or

(b) What are the goals of task analysis? Explain.

14. (a) Explain black-box and white-box testing methods.

Or

(b) What are the test approaches used for client-server applications?

15. (a) Mention the goals of risk management.

Or

(b) Write a note on ISO 9000 quality standards.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the key characteristics of a software.

17. Discuss the key elements used to build an analysis model.

18. Explain the various types involved in the architectural design.

19. Discuss the testing strategies used for object oriented software.

20. Discuss different types of risks and their identification methods.

RW–10488

Sub. Code
541301/ 545301

M.C.A. (R) /M.C.A.(WE) DEGREE EXAMINATION, NOVEMBER 2015

COMPUTER GRAPHICS

(Common for M.C.A.(R)/M.C.A.(WE))

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. List out the applications of computer graphics.

2. What is ellipse algorithm?

3. What is the purpose of homogeneous coordinates?

4. Differentiate between 2D and 3D geometric transformation.

5. Write down the properties of Bezier curve.

6. List out the steps used in 3D transformation.

7. What is rotation?

8. What do mean by viewing pipeline?

9. What is color look up table?

10. Define tweening.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

All questions carry equal marks.

11. (a) Discuss about raster scan system.

Or

(b) Write the steps in midpoint circle algorithm.

12. (a) Write a note on transformation.

Or

(b) Explain about clipping operation.

13. (a) Differentiate between interpolation spline and approximation spline.

Or

(b) Write a note on polygon rendering methods.

14. (a) Write the general steps on three dimensional rotation.

Or

(b) What is projection transform? Explain with suitable example.

15. (a) Explain in detail about depth sorting method.

Or

(b) Write in detail about key frame systems.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

All questions carry equal marks.

16. Discuss in detail about boundary fill algorithm.

17. Explain Sutherland Hodgeman polygon clipping algorithm with an example.

18. Explain in detail about Bezier curves and surface.

19. Discuss about reflection and shear transformation.

20. Write a detailed note on back face detection method.

———————

RW–10489

Sub. Code
541302/ 545302

M.C.A. (R)/WEP. DEGREE EXAMINATION, NOVEMBER 2015

Computer Application

INTERNET AND JAVA PROGRAMMING

(2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. Define domain name system.

2. How the user can chat in online system?

3. List out the basic concepts of OOP.

4. Write notes on constant variable.

5. Define class with suitable example.

6. What is an array? How to create an array?

7. How to create a thread in class? Give an example.

8. Write the syntax of an exception handling code.

9. List out the stream classes in file.

10. How to create a file?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

All questions carry equal marks.

11. (a) What is Internet and write about the resources of Internet?

Or

(b) Explain about FTP and TELNET.

12. (a) Write the features of java programming.

Or

(b) How is java associated strongly with Internet?

13. (a) Describe the nesting of methods with an example.

Or

(b) Define an interface. Write the difference between class and interface.

14. (a) How can you extend the thread class?

Or

(b) Discuss about the types of error in java.

15. (a) Discuss about the character scheme classes.

Or

(b) Write short note on reading and writing a character.

Part C (3 × 10 = 30)

Answer any **three** questions.

All questions carry equal marks.

16. What is the usage of web? Describe the IRC and UseNet newsgroup.

17. Describe the classification of operators used in java.

18. Describe the various forms of implementing interface and give an example.

19. Explain how can use exception for debugging.

20. Explain the input and output exception with suitable example.

———————

RW–10490

Sub. Code
541303/ 545303

M.C.A. DEGREE EXAMINATION, NOVEMBER 2015

COMPUTER NETWORKS

(Common for M.C.A. (R)/M.C.A.(WEP))

(2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. Define multicasting.

2. What is protocol?

3. Define frame.

4. Define bit stuffing.

5. What is sink tree?

6. Define reverse path forwarding.

7. What is SMTP?

8. Define Hyperlinks.

9. What is symmetric key cryptography?

10. What is mono alphabetic substitution?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Discuss about TCP and UDP.

Or

(b) Explain Twisted pair cable.

12. (a) Explain cyclic redundancy check.

Or

(b) Explain CSMA/CD.

13. (a) What is circuit switching? Explain.

Or

(b) Explain congestion prevention policies.

14. (a) Write about UDP.

Or

(b) Write about domain name system.

15. (a) Explain substitution cipher techniques in cryptography.

Or

(b) Explain triple DES.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

All questions carry equal marks.

16. Draw the OSI reference model and explain.

17. Explain sliding window protocols in detail.

18. Explain distance vector routing.

19. Describe the functionalities of transport layer.

20. Explain RSA cryptographic technique with an example.

———————

RW–10491

Sub. Code
541551/ 545551

M.C.A. (R)/M.C.A. (WE) DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

OBJECT ORIENTED ANALYSIS AND DESIGN

(Common for M.C.A. (R)/M.C.A. (WE))

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Why is encapsulation important?

2. What is the difference between an object's methods and an object's attribute?

3. What are the contents of UML behaviour diagrams?

4. List the five Booch diagrams.

5. How would you name classes?

6. What is a use case?

7. What is the relationship between coupling and cohesion?

8. What are the tasks of Access Layer class?

9. Define Multiplicity attribute.

10. What is client/server technology?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Write a short note on kinds of relationships.

Or

(b) What is Cardinality? Explain its types.

12. (a) Briefly describe the Booch system development process.

Or

(b) Explain the forms of associations.

13. (a) List the guidelines for identifying super – sub relationships.

Or

(b) Explain the steps for finding usecases.

14. (a) List out the difference between good design and bad design.

Or

(b) Explain the steps for view layer macro process.

15. (a) Write a short note on client/server application's component.

Or

(b) List the characteristics of client/server architecture.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Discuss the object oriented system development in detail.

17. Explain the Rumbaugh object modeling technique.

18. Discuss the Use – Case model with an example.

19. Describe the database interfaces in detail.

20. Describe the application of Object Oriented technology to the creation of a foundation class library.

———————

RW–10492

Sub. Code
541560 / 545560

M.C.A. DEGREE EXAMINATION, NOVEMBER 2015

Mathematics

RESOURCE MANAGEMENT TECHNIQUES

(Common for MCACR/MCA (WEP))

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. What is meant by unbounded solution?

2. What is a feasibility region?

3. How do you detect an unbounded solution in the simplex procedure?

4. Give the mathematical formulation of an assignment problem.

5. What is the need of Integer Programming problem?

6. What do you mean by project?

7. Differentiate between PERT and CPM?

8. What is float? What are different types of floats?

9. What is joking in Queuing theory?

10. What is the average waiting time in a queue and average queue length in (M/M/1) model?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Use the two phase simplex method to

Maximize $z = 5x_1 - 4x_2 + 3x_3$

Subject to

$$\begin{aligned} 2x_1 + x_2 - 6x_3 &= 20 \\ 6x_1 + 5x_2 + 10x_3 &\leq 76 \\ 8x_1 - 3x_2 + 6x_3 &\leq 50 \\ x_1, x_2, x_3 &\geq 0. \end{aligned}$$

Or

(b) Write about the basic principles of simplex method and two-phase method with an example.

12. (a) Solve the LPP using cutting plane method :

Maximize $Z = x_1 + x_2 + x_3$

Subject to

$$\begin{aligned} 7x_1 - 5x_2 + x_3 &\leq 7 \\ -12x_1 + 15x_2 + x_3 &\leq 7 \\ x_1, x_2, x_3 &\geq 0 \end{aligned}$$

and integer.

Or

(b) List out the differences between the transportation problem and the assignment problem.

13. (a) Define integer programming problem and list the various types of integer programming problem.

Or

(b) Write the characteristics of the standard form for linear programming problem.

14. (a) Explain about Crashing of Networks.

Or

(b) Discuss the methods of project scheduling by PERT.

15. (a) Explain various queuing with example.

Or

(b) Write a note on inventory management.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Use Penalty method to solve :

Minimize $Z = 4x + 3y$

Subject to the constraints :

$$2x + y \geq 10,$$
$$2y - 3x \leq 6,$$
$$x + y \geq 6,$$

and $x \geq 0, y \geq 0$.

17. A company has three production facilities, S_1, S_2, S_3 with production capacity 7, 9 and 18 units per week of a product, respectively. These units are to be shipped to the four warehouses D_1, D_2, D_3 and D_4 with requirements of 5, 8, 7 and 14 units per week, respectively. The transportation cost (in rupees) per unit between factories and warehouses are given below :

	D_1	D_2	D_3	D_4
S_1	19	30	50	10
S_2	70	30	40	60
S_3	40	8	70	20

Find the optimal solution of the transportation problem.

18. Find the optimum integer solution to the following IPP by using cutting plane method :

Maximize $z = 5x_1 + 8x_2$.

Subject to $x_1 + 2x_2 \leq 8$

$4x_1 + x_2 \leq 10$

$x_1, x_2 \geq 0$ and integers.

19. The time estimate (in weeks) for the activities of a PERT network are given below :

	Duration		
Job (I, j)	Optimistic (t_o)	Most likely (t_m)	Pessimistic (t_p)
1-2	1	1	7
1-3	1	4	7
1-4	2	2	8
2-5	1	1	1
3-5	2	5	14
4-6	2	5	8
5-6	3	6	15

(a) Determine the expected project length.

(b) If the project due date is 19 weeks, what is the probability of not meeting the due date?

20. Show that the average number of units in a (M/M/I) queuing system is equal to $\rho(1 - \rho)$.

RW–10493

Sub. Code
541501/ 545501

M.C.A. (R)/M.C.A. (WE) DEGREE EXAMINATION, NOVEMBER 2015

Fifth Semester

MULTIMEDIA AND ITS APPLICATIONS

(Common for M.C.A. (R)/M.C.A. (WEP))

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Write the difference between multimedia and hypermedia.

2. List out the various audio file formats.

3. What are the advantages of digital video over analogue video?

4. What is image scanner?

5. Compare variable length coding and run length coding.

6. What is compression ratio?

7. What is the need of data compression?

8. Write down the techniques used in audio compression.

9. What is multimedia?

10. What is MOD?

Part B $(5 \times 5 = 25)$

Answer **all** the questions, choosing either (a) or (b).

All questions carry equal marks.

11. (a) List out the important issues in multimedia system.

Or

(b) Write a brief note on color models in images.

12. (a) Compare and contrast analog video and digital video.

Or

(b) Write the steps followed in lossy compression algorithm.

13. (a) Describe the run-length coding algorithm.

Or

(b) Explain about quantization.

14. (a) Discuss about audio compression techniques.

Or

(b) List out the difficulties in video compression.

15. (a) Write a detailed note on multimedia network communications.

Or

(b) Explain multimedia over ATM networks.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

All questions carry equal marks.

16. Discuss in detail about multimedia applications in real world.

17. Write a detailed note on MIDI.

18. Explain the Arithmetic coding algorithm.

19. Explain in detail about MPEG compression techniques.

20. Explain the quality of multimedia data transmission.

RW–10494

Sub. Code
541502/ 545502

M.C.A. DEGREE EXAMINATION, NOVEMBER 2015

DATA MINING AND WAREHOUSING

(Common for M.C.A. (R)/M.C.A. (WEP))

(2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** the questions.

1. How is a data warehouse differ from data mining?

2. Write down the applications of data warehousing.

3. What is the aim of data mining?

4. Define Data cleaning.

5. Define Association Rule Mining.

6. What is Decision tree?

7. What are the uses of statistics in data mining?

8. Define pattern evaluation.

9. What does web mining mean?

10. Define spatial data mining.

Section B (5 × 5 = 25)

Answer **all** the questions.

11. (a) Describe the Warehouse Server Models.

Or

(b) Explain how data mining is used in Tourism Industry.

12. (a) Explain the different forms of knowledge.

Or

(b) Write a short note on visualization.

13. (a) Explain Apriori algorithm.

Or

(b) Briefly explain the back propagation method.

14. (a) Write a short note on working principles of GA.

Or

(b) Briefly explain the K-Means algorithm.

15. (a) Explain Temporal mining.

Or

(b) Explain the mining of time series and sequence data.

Section C (3 × 10 = 30)

Answer any **three** questions.

16. Describe the OLAP operations.

17. What are the types of data pre-processing techniques? Explain in detail about them.

18. Explain the various classification methods.

19. Discuss the Clustering algorithms in detail.

20. Explain the various mining tools in detail.

———————

RW–10495

Sub. Code
541503/ 545503

M.C.A. DEGREE EXAMINATION, NOVEMBER 2015

MOBILE COMMUNICATION

(Common for M.C.A. (R) / M.C.A. (WEP))

(2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is multipath propagation?

2. What is necessity of guard space?

3. Specify the security services offered by GSM

4. Define: Handover

5. What is the difference between infrastructure and ad-hoc networks?

6. What is the goal of WLANs?

7. Define a tunnel

8. What are the requirements of mobile IP?

9. What is slow start?

10. What is the use of M-TCP?

Part B $(5 \times 5 = 25)$

Answer any **all** questions.

11. (a) List the difference between SDMA/TDMA/FDMA/CDMA.

Or

(b) Explain the spread spectrum in detail.

12. (a) Write a short note on mechanism of GPRS.

Or

(b) Explain the control channel groups of GSM.

13. (a) List the advantages of WLANs .

Or

(b) What are advantages and problems of forwarding mechanisms in Bluetooth networks?

14. (a) Explain the entities of mobile IP.

Or

(b) Write a short note on encapsulation of mobile IP.

15. (a) List the features of WML.

Or

(b) State the layers in WAP architecture.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. (a) Explain about cellular wireless network.

(b) Explain the wireless transmission

17. Describe the GSM architecture in detail.

18. Describe the system architecture of IEEE 802.11.

19. Describe Dynamic host configuration protocol.

20. Explain the followings:
    (a) Indirect TCP
    (b) Snooping TCP
    (c) Mobile TCP

RW–10496

Sub. Code
541553/ 545553

M.C.A. ((R)/WE) DEGREE EXAMINATION, NOVEMBER 2015

Fifth Semester

Computer Applications

SOFTWARE TESTING METHODOLOGIES

(Common for M.C.A. (R)/M.C.A. (WEP))

(2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define BUGS.
2. What is meant by path testing?
3. Define interface testing.
4. Mention the basics of data flow testing.
5. Define path.
6. Give the purpose of constructing decision tables.
7. What is meant by state testing?
8. How to determine the power of a matrix?
9. How to create the test script?
10. Give the expansion for HTTP.

Part B $(5 \times 5 = 25)$

Answer **all** questions by choosing either (a) or (b).

11. (a) Give the purpose of testing.

Or

(b) How to construct flow graphs?

12. (a) Discuss the strategies involved in data flow testing.

Or

(b) Explain the concept of domain testing.

13. (a) How to detect the flow anomaly?

Or

(b) How to construct kr charts?

14. (a) When state graphs are said to be good and bad?

Or

(b) How to determine the matrix of a graph?

15. (a) Discuss the usage of Jmeter.

Or

(b) How to synchronize the test case?

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Discuss on the applications of path testing.

17. Explain the techniques involved in transaction flow testing.

18. How to perform logic based testing?

19. Describe the node reduction algorithm.

20. Discuss the win runner tools for functional testing.

———————

RW–10497

Sub. Code
541562/ 545562

M.C.A. DEGREE EXAMINATION, NOVEMBER 2015

Computer Applications

GRID COMPUTING

(Common for (MCA (R)/ MCA (WEP))

(2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is grid computing?
2. What are the types of grids?
3. Classify the grid tools.
4. What is the advantages of grid computing over distributed computing?
5. What are the components supported by SNIA shared storage model?
6. Write the characteristics of web service.
7. What are the benefits provided by OGSI standard?
8. Write the features of SOAP.
9. What are the grid services required for program execution?
10. Give any two issues that arise in data caching.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Describe the components of grid computing systems.

Or

(b) Compare virtualization with grid computing.

12. (a) Give a brief history of communications.

Or

(b) Write the challenges of grid computing.

13. (a) Describe the functional view of grid computing.

Or

(b) What are the specifications developed for performing grid computing on top of web services? Explain them.

14. (a) Explain the protocol – oriented view of a grid architecture.

Or

(b) Write the objectives of OGSA.

15. (a) Describe the grid services included in resource management category.

Or

(b) Write about common management model.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. With a neat diagram, explain the grid computing environment.

17. Explain the study undertaken to evaluate the economic value of the introduction of grid computing.

18. Describe the physical view of grid resources.

19. What are the logical components of OGSA? Explain them.

20. Describe the system properties requirements in detail.

RW–10632

Sub. Code
541201/ 545201

M.C.A. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

OBJECT ORIENTED PROGRAMMING AND C++

(Common for M.C.A. (R)/WE)

(2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

All questions carry equal marks.

1. What are the tokens in C++?
2. Define class.
3. Write out the syntax of declaring function prototyping.
4. How to declare call by reference in function?
5. Define inheritance.
6. Define polymorphism.
7. What is class template?
8. Write a syntax of open and close a file.
9. Write a short note on exception handling.
10. What is exception in class templates?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain about the basic concepts of object oriented programming.

Or

(b) Describe about the hierarchy of console stream classes.

12. (a) Describe about the array of object with suitable example.

Or

(b) Define constructor. Describe the special characteristics of constructor.

13. (a) Explain about multiple Inheritance with suitable example.

Or

(b) Describe about the virtual base class.

14. (a) Write short notes on class template with multiple arguments with suitable example.

Or

(b) Describe the hierarchy of file stream classes.

15. (a) Describe the Exception in constructor and destructor.

Or

(b) Explain about the memory allocation failure exception.

Part C (3 × 10 = 30)

Answer any **three** questions.

All questions carry equal marks

16. Explain about the formatted console I/O operation.

17. What is friend function? Explain with an example.

18. Discuss about operator overloading with suitable example.

19. Explain the file pointer and their manipulators.

20. Illustrate Exception Handling Mechanism with an example and explain.

———————

RW–10633

Sub. Code
541202/ 545202

M.C.A. (R/WE) DEGREE EXAMINATION, NOVEMBER 2015

OPERATING SYSTEMS

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define Operating System.

2. What do you mean by tightly coupled system?

3. What is a Process?

4. What is the function of CPU scheduler?

5. What is Race condition?

6. Write the conditions for Deadlock.

7. What is meant by Internal Fragmentation?

8. What is meant by Segmentation?

9. What are the attributes of a File?

10. Write the features of Linux.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Explain various system programs.

Or

(b) Explain the structure of operating system.

12. (a) Explain the various operations on process.

Or

(b) Explain inter process communication.

13. (a) Explain the methods for handling Deadlock.

Or

(b) Explain the conditions for Deadlock with example.

14. (a) Explain any one page replacement algorithm with an example.

Or

(b) Explain various File access methods.

15. (a) Explain the file system implementation with diagram.

Or

(b) Discuss free space management.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain various operating system services.

17. A system has following jobs to generate with a processor

Job :	1	2	3	4
Duration :	7	4	1	5
Arrival time :	0	2	4	5

The jobs have arrived in the order 1,2,3,4. Find the waiting time and Turnaround time of each job for the following algorithms.

(a) Preemptive Shortest Job First (SJF).

(b) Round Robin with a time slice of one unit.

18. What is Semaphore? Explain with code how it can be used to solve critical section problem.

19. Explain the Directory structure with a diagram.

20. Explain :

(a) Disk Scheduling

(b) Disk Management.

RW–10634

Sub. Code
541204/ 545204/ 551203

M.C.A. DEGREE EXAMINATION, NOVEMBER 2015

COMMUNICATION SKILLS

(Common for M.C.A., R/WE/M.Sc., Computer Science)

(2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** the questions.

1. Define intrapersonal communication.

2. What communication takes place when people working at the same level interact with those working at a higher or lower-level of organisational hierarchy and across the boundaries of their reporting relationships?

3. What is the mode of apologizing to a person?

4. How would you formally express disagreement?

5. What is chronemics?

6. What are audio-visual aids?

7. What are the essential qualities of a good leader?

8. Why do we conduct meeting?

9. What are the essential parts of minutes?

10. Define resumé.

Section B (5 × 5 = 25)

Answer each of the following questions, choosing either (a) or (b).

11. (a) What are the advantages and limitations of upward communication?

Or

(b) Bring out the different levels of communication.

12. (a) Explain the various modes of introduction.

Or

(b) What are the attributes of soft skills?

13. (a) Analyse the significance of planning.

Or

(b) Highlight the advantages of using audio-visual aids in presentation.

14. (a) Explain patterns of behavior in Group Communication.

Or

(b) Bring out the importance of team work.

15. (a) How would you prepare an agenda?

Or

(b) Discuss the advantages of e-mail.

Section C (3 × 10 = 30)

Answer any **three** questions.

16. Examine the different levels of communication.

17. Discuss the various modes of conversation.

18. Bring out the nuances involved in presentation.

19. Chart out the chief features of Group Discussion.

20. Examine the components included in Curriculum Vitae.

RW–10635

Sub. Code
541205/ 545205

M.C.A. DEGREE EXAMINATION, NOVEMBER 2015

ACCOUNTING AND FINANCIAL MANAGEMENT

(Common for M.C.A. (R)/(WE))

(2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A $(10 \times 2 = 20)$

Answer **all** the questions.

1. What is going concern concept?

2. State any two advantages of trial balance.

3. What is BEP?

4. What is cost sheet?

5. What is labour variance?

6. What are the uses of Budgets?

7. What are the advantages of preparing cash budget?

8. What is IRR?

9. What is gross working capital?

10. What is cost of debt?

Section B (5 × 5 = 25)

Answer **all** the questions.

11. (a) Prepare Trading Account of Archana for the year ending 31.12.14 from the following information.

	Rs.		Rs.
Opening stock	1,80,000	Purchase returns	10,000
Purchases	8,60,000	Sales returns	3,16,000
Freight inward	52,000	Closing stock	1,00,000
Wages	24,000	Import duty	30,000
Sales	14,40,000		

Or

(b) What are the uses of cash flow statement?

12. (a) On the basis of following information, calculate

(i) G/p ratio

(ii) Working capital turnover ratio

(iii) Debt-equity ratio

Net sales	30,00,000
Cost of goods sold	20,00,000
Current assets	6,00,000
Current liabilities	2,00,000
Paid up share capital	5,00,000
Debentures	2,50,000
Loan	1,25,000

Or

(b) What is marginal costing? Why it is important?

13. (a) Discuss the components of budgetary control system.

Or

(b) A factory is currently running at 50% capacity and produces 5,000 units at a cost of 900 per unit as per details below:

Material Rs. 500, Labour Rs. 150, Factory overheads Rs. 150 (60 fixed), Administrative overheads Rs. l00 (50 fixed), The current selling price is 1,000 per unit. At 70% working, material cost per unit increases by 2% and selling price per unit falls by 2%. Estimate profits of the factory at 70% working by preparing a flexible budget.

14. (a) Discuss the different methods of evaluating a capital expenditure proposal. Give the features of each method.

Or

(b) From the following information calculate post payback profitability

Initial investment	Rs. 25,000
Annual cash inflow	5,000
Life	6 Years
Payback period	4 Years

15. (a) What is Modigliani and Miller (MM) theory? Explain briefly.

Or

(b) Velvet Corporation needs to issue new long-term debt in order to raise capital. Their existing long-term bonds have various coupons, but currently have a YTM of approximately 10%. Therefore Velvet believes they could issue new, 20-year annual-pay bonds with a 10% coupon. Velvet will incur a $20 flotation cost an each bond sold, meaning they will net only $ 980 per bond. Velvet has a corporate tax rate of 40%. What is the/after-tax cost of debt for Velvet Corporation?

Section C

(3 × 10 = 30)

Answer any **three** questions.

16. Distinguish between fund flow statement and cash flow statement.

17. Describe are the important decision making areas of Marginal costing.

18. A company is to start a new project which is having cost of Rs. 50,000/- and life of 5 years. Salvage value is nil, tax rate for the company is 55% and it follows Straight line method of depreciation. The cash flows before tax (CFBT) are as follows:

Year :	1	2	3	4	5
CFBT (Rs.)	10,000	11,000	20,000	30,000	35,000

Using 10% cost of capital, compute the following :

(a) Payback period.

(b) Average rate of return.

(c) NPV at 10 % discount factor.

19. Discuss the requirements of sound working capital management.

20. Mr. Amol, manufacturer of an electronic product, sells his product at Rs. 10/- each, variable cost being Rs. 4/- per unit and fixed cost amounting to Rs. 60,000/-.

(a) Calculate the Break Even Point.

(b) What will be the profit if Mr. Amol sells 30,000 units?

(c) What will be the BEP, if Mr. Amol spends Rs.6000/- on advertising?

RW–10636

Sub. Code
541402/ 545402

M.C.A. DEGREE EXAMINATION, NOVEMBER 2015

DISTRIBUTED COMPUTING

(Common for M.C.A. (R/WE))

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. Write an example for distributed system.

2. List out the various types of networks.

3. What are the issues in group communication?

4. What is remote invocation?

5. Mention the uses of digital signature.

6. What is the security mechanism used in distributed computing?

7. List out the modes of synchronization.

8. What are the different algorithms for election?

9. What is dirty read?

10. Define Deadlock.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

All questions carry equal marks.

11. (a) Explain the various types of system models.

Or

(b) Mentions the networking issues in distributed systems.

12. (a) Explain the characteristics of inter process communication.

Or

(b) Write a note on java RMI.

13. (a) What is thread? Explain its types.

Or

(b) Write a brief note on flat file service interface.

14. (a) Mention the steps for updating the clocks.

Or

(b) Explain bully algorithm.

15. (a) Explain the rules to lock nested transaction.

Or

(b) Write a note on distributed multimedia system.

Part C (3 × 10 = 30)

Answer any **three** questions.

All questions carry equal marks.

16. Discuss about the challenges in distributed systems.

17. Illustrate the client server communication and explain.

18. Explain about sun network file system.

19. Write explanation on logical time and logical clocks.

20. Explain in detail about transaction recovery system.

———————

RW–10637

Sub. Code
541403/ 545403

M.C.A. DEGREE EXAMINATION, NOVEMBER 2015

WEB TECHNOLOGY

(Common for M.C.A. (R)/WE)

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. Difference between HTML and XHTML.
2. What is XML namespace?
3. List out the advantages of Java Beans.
4. What are cookies?
5. Compare and contrast POST & GET methods
6. What are the tasks of servelets?
7. List the components to built an web server
8. Difference between JSP and ASP.
9. What are the types of JSP directories?
10. Write the issues on Apache web server.

Part B $(5 \times 5 = 25)$

Answer ALL questions, choosing either (a) or (b).

All questions carry equal marks.

11. (a) Explain the use of meta elements in XHTML with suitable example.

Or

(b) List the advantages of using XML.

12. (a) What is JavaBeans component?

Or

(b) Mentionl the features of EJB.

13. (a) Briefly explain HTTP request methods.

Or

(b) Explain about session tracking and management.

14. (a) Design an online shopping using JSP script.

Or

(b) Explain session handling injava servelets.

15. (a) Briefly explain the SOAP elements.

Or

(b) Mention the uses of java bean classes in JSP.

Part C (3 × 10 = 30)

Answer any **three** questions.

All questions carry equal marks.

16. Explain dynamic webpage creation for student mark sheet generation.

17. Discuss about the relationship between Java Bean and EJB.

18. Explain about the life cycle of serverlet.

19. Explain the creation of java web services with an example.

20. Explain JDBC connectivity in detail.

———————

RW–10483

Sub. Code
541101/ 545101

M.C.A. DEGREE EXAMINATION, NOVEMBER 2015

DIGITAL COMPUTER ORGANIZATION

(Common for MCA (R)/MCA (WEP))

(2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. State De Morgan's Theorem.

2. What is the use of K-Map?

3. Define Full-adder with Circuit.

4. What is the Purpose of Shift Register?

5. Define Computer Register.

6. List the uses of accumulator.

7. Define Stack.

8. Define Addressing Modes.

9. What is the functionality of DMA?

10. What is Auxiliary Memory?

Part B $(5 \times 5 = 25)$

Answer **all** the questions.

11. (a) Explain the procedure of Octal to Hexadecimal Conversion with Example.

Or

(b) Explain Product of Sum with block diagram.

12. (a) Explain Multiplexers with its circuit.

Or

(b) Briefly explain the Error detection codes.

13. (a) Explain Timing and Control in detail.

Or

(b) Explain the memory reference instructions.

14. (a) Write a short note on Addressing modes with examples.

Or

(b) Explain Stack Organisation.

15. (a) Explain Asynchronous Data Transfer.

Or

(b) Explain Associative Memory in detail.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Discuss in detail on Karnaugh map simplification with an example.

17. Illustrate the different type of Combinational Circuits and explain.

18. Explain design of accumulator logic with block diagram

19. Discuss :

    (a) Instruction format with examples

    (b) Program Control

20. Explain the types of Memory organisation in detail.